

Influenza vaccination and reduction in risk of ischemic heart disease among chronic obstructive pulmonary elderly

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ARTICLE INFO

Article history:

Received 8 March 2013

Received in revised form

22 April 2013

Accepted 15 May 2013

Keywords:

Influenza vaccine

COPD (chronic obstructive pulmonary disease)

IHD (ischemic heart disease)

Vaccination

Cohort research database

ABSTRACT

COPD estimates the third leading cause of death by 2020. To reduce its complications becomes an important issue. The purpose of this study is to investigate the influenza vaccine effect for IHD occurrence with secondary to COPD. We used 11 years Taiwan National Health Insurance cohort research database and analyzed the relationships between vaccination and incident of IHD for COPD patients stratified by age. Total 29,178 COPD patients, 703 patients got vaccination out of 1464 who have developed IHD and 6010 patients vaccinated out of 27,714 that did not developed IHD. Major findings we observed: vaccination was associated with a reduced risk of IHD (OR, 0.746; 95% CI, 0.595–0.937) in elderly COPD patients. Influenza vaccination was associated with a reduced risk for IHD only in elderly COPD patients. Moreover, COPD patients with IHD had higher vaccination rate than without IHD.

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1. Introduction

Chronic obstructive pulmonary disease (COPD) is a serious global health concern and a major cause of chronic morbidity and mortality worldwide. WHO (World Health Organization) estimates COPD is the fourth leading cause of death and projected to be the third by 2020 [1,2]. As chronic obstructive pulmonary disease could be preventable and managed in this era. Influenza vaccinations currently recommended in the care of people with COPD [3,4]. Since COPD is within

the top 10 leading cause of deaths in Taiwan from many years, and the prevalence rate has reached to 6.7%. Many of COPD patients die due to cardiovascular disease [5,6]. Moreover, the important is how to reduce its complications and how to do its treatment [7]. While the ischemic heart disease is the most common concurrent disease with COPD, therefore to find an intervention, which can reduce heart disease, is also an important issue. Previous studies have shown that influenza vaccination can reduce hospitalization of patients with heart and lungs diseases [6,8–11]. As American Heart Association indicated, guidelines in secondary prevention

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<http://dx.doi.org/10.1016/j.cmpb.2013.05.006>

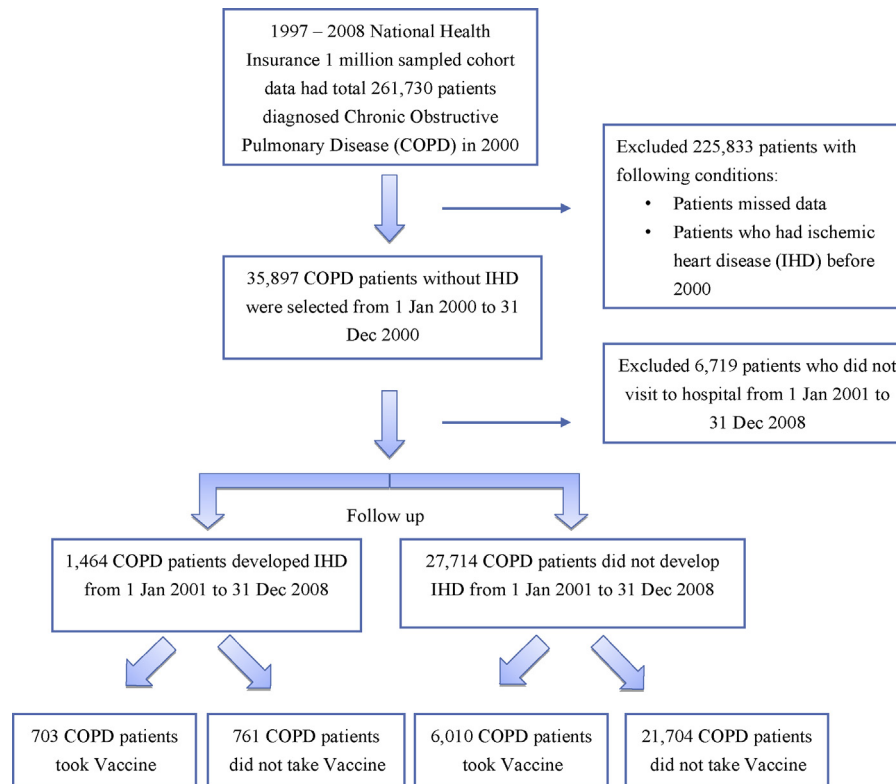


Fig. 1 – Flow chart to identify the patients through database.

indicate that patients with cardiovascular disease should have an influenza vaccination [12,13]. We conducted a population-based study in Taiwan by using Bureau of National Health Insurance (BNHI) research database to investigate the outcome of annual influenza vaccination on incidence for IHD among COPD patients.

2. Methods

The Bureau of National Health Insurance program, started in March 1995, covers 96% of Taiwanese population. The Department of Health (DOH) of Taiwan started to sponsor influenza vaccine for high-risk elderly patients in 1998. Influenza vaccination has been free of charge since 2001 to every elderly person over the age of 65 [6]. We used one-million cohort sample of BNHI data from 1997 to 2008. The total 261,730 patients were selected in 2000 that diagnosed with COPD including asthma (ICD-9-CM International Classification of Diseases, ninth revision, Clinical Modification Codes: 491.xx, 492.xx, 493.xx and 496) and not have had diagnosed with ischemic heart disease (IHD codes: 410.xx and 411.xx) in the last three years. We excluded 225,833 patients with missing data or have had IHD diagnosis in past three years. After inclusion of 35,897 patients, 6719 patients did not visit the hospital from 2001 to 2008 that were also excluded from our study. Finally, we pursued with 29,178 patients and used only the first recorded episode of COPD diagnosis. We identified through the patient medical records occurring in 2000 that enrolled and followed until 2008 whether they developed IHD while being vaccinated with influenza vaccination or not.

We used Microsoft SQL Server 2008 and SPSS 18 for computation and analysis. The descriptive statistics analysis used for distributions of variables and ANOVA, Chi-square test (χ^2 -test) for exploring the relationships between variables. The odds ratios (OR) and 95% confidence intervals (CI) were estimated and two-sided *p*-value of less than 0.05 was considered statistically significant for this study.

3. Results

During the eight year follow-up period, among the total 29,178 COPD patients, 703 patients get vaccination out of 1464 who have developed ischemic heart disease (vaccination rate 48.02%) and 6010 patients got vaccinated out of 27,714 who did not developed ischemic heart disease (vaccination rate 21.69%). The flow of patients throughout the whole study has shown in Fig. 1.

After identification a total of 29,178 including 14,581 males out of that 876 (59.8%) developed IHD and 13,705 (49.5%) did not develop IHD, 14,593 were females out of that 584 (40.2%) developed IHD and 14,009 (50.5%) did not develop IHD. The overall common risk for IHD patients for hypertension is 77.2%, diabetes mellitus is 40.4% and lipid disorders is 49.2% while comparing with non-IHD patients having hypertension is 41.6%, diabetes mellitus is 26.5% and lipid disorders is 36.3% (Table 1).

In this study, we also compared with non-COPD patients to stratify our results and found that IHD incident rate for COPD patients is about four folds then non-COPD (0.18% and 0.58%, respectively). The vaccination rate of ischemic heart

Table 1 – Characteristic of COPD patients.

Characteristic	Ischemic heart disease (N = 1464)	Non ischemic heart disease (N = 27,714)	p-Value
Gender			<0.001
Male (%)	876 (59.8)	13,705 (49.5)	
Female (%)	584 (40.2)	14,009 (50.5)	
Overall common risk			
Hypertension (%)	1130 (77.2)	11,518 (41.6)	<0.001
Diabetes (%)	596 (40.7)	7348 (26.5)	<0.001
Lipid disorders (%)	720 (49.2)	10,046 (36.3)	<0.001
Common risk for vaccinated (%)			
Hypertension (%)	703 (48)	6010 (21.7)	<0.001
Diabetes (%)	580 (82.5)	4754 (79.1)	<0.001
Lipid disorders (%)	303 (43.1)	2807 (46.7)	<0.001
	334 (47.5)	3162 (52.6)	<0.001

disease patients among non-COPD versus COPD is 25.15% versus 48.02% (Table 2). We tried to understand the effect of vaccine on COPD with IHD differentiated by stratifying the age. The COPD patients that developed IHD younger than 40 years was about 3.8% of the whole IHD population therefore we excluded them and analyzed the patients by every 10 years decade above 40 years by using Chi-square test.

In the Chi-square test analysis, influenza vaccination was associated with a reduced risk of IHD (OR, 0.746; 95% CI, 0.595–0.937) in elderly COPD patients in the age group of 71–80 years old. Although the prevalence history of hypertension, diabetes, dyslipidemia in COPD patients was found higher in IHD developed group than the non-IHD developed group (Table 3).

4. Discussion

In this population-based study, we demonstrated that COPD patients have associated decreased risk of IHD while vaccinated with influenza vaccine over many years [14]. Furthermore, the influenza vaccination provided significant reduction of ischemic heart disease events in elderly, suggesting that the influenza vaccine could provide effective protection of IHD in COPD patients especially in elderly group.

It is also very common to find comorbidities within the first year of diagnosis. Soriano et al. found many other conditions with frequencies of at least 1% at the time of COPD diagnosis which is the highest being angina with a prevalence of 4.0% [15]. Moderate COPD risk of cardiovascular disease

in non-COPD patients is 2-folds [16,17]. Despite the almost universal recommendation that people with chronic obstructive pulmonary disease (COPD) should receive an annual influenza vaccination, there are no trials have yet been evaluated the effect of influenza vaccination in the incidence of these patients with comorbidities [18]. A meta-analysis studies showed slight trend in preventing mortality, which indicating that influenza vaccination impact needs to be further investigated [19].

In Taiwan, the free influenza vaccination program began in October 1999 targeted at the high-risk elderly population and later on it was recommended for all adults above 65 years old in Taiwan since 2001 [6]. Some researchers found that influenza vaccination might reduce the occurrence of IHD during influenza season [9,20]. Although, the mechanism of influenza infection exerts on ischemic heart disease remains unclear. However, in this population-based study, we tried to investigate the effect of influenza vaccination for COPD patients having ischemic heart disease (IHD secondary to COPD patients) which has clinically no direct relationship with influenza vaccine.

However, our study investigated the influenza vaccine association with ischemic heart disease but there are still some limitations to our study. Firstly, the diagnosis of disease COPD, IHD and vaccination status were identified on the basis of the ICD-9 codes or drug codes, in which the diagnosis made exclusively by certified physicians for insurance claims which did not checked externally.

Secondly, we also assumed that the patient took influenza vaccination at least once during our study period identified by

Table 2 – The incident rate of IHD comparison between COPD and non-COPD groups.

Year	COPD patients		Non-COPD patients	
	IHD incident rate	Vaccination rate	IHD incident rate	Vaccination rate
2001	0.86%	43.56%	0.17%	21.40%
2002	0.59%	52.82%	0.17%	23.50%
2003	0.56%	50.00%	0.17%	24.41%
2004	0.56%	52.81%	0.18%	26.44%
2005	0.57%	45.51%	0.19%	25.11%
2006	0.53%	43.40%	0.17%	27.54%
2007	0.50%	53.42%	0.19%	26.07%
2008	0.44%	44.72%	0.18%	27.71%
All	0.58%	48.02%	0.18%	25.15%

Table 3 – COPD patients who developed IHD and who did not developed IHD.

Age group	Ischemic heart disease		Non ischemic heart disease		Odds ratio (95% CI)
	Total	Vaccinated (%)	Total	Vaccinated (%)	
0–10	4	0 (0)	3133	11 (0.4)	–
11–20	3	0 (0)	1837	3 (0.2)	–
21–30	12	0 (0)	2304	12 (0.5)	–
31–40	37	0 (0)	3850	30 (0.8)	–
41–50	135	1 (0.7)	4514	57 (1.3)	0.58 (0.03–3.94)
51–60	225	9 (4.0)	3880	210 (5.4)	0.73 (0.35–1.48)
61–70	490	287 (58.6)	4517	2804 (62.1)	0.86 (0.71–1.05)
71–80	465	340 (73.1)	3270	2566 (78.5)	0.75* (0.60–0.93)
81–90	92	65 (70.7)	406	314 (77.3)	0.71 (0.41–1.21)
91–100	1	1 (100)	3	3 (100)	–

* p-Value < 0.01.

patient medical records. Personal information, such as body weight, smoking habits, and biochemistry profiles, were not available in the datasets [21,22]. Finally, we did not discuss the severity of COPD disease because it is not easy to identify in a retrospective database. However, recently a study done by using the claims data to stratify the complexity of patients [23]. There may be need further investigations or clinical trials needed on different population samples.

5. Conclusion

The findings showed that influenza vaccination was associated with a reduced risk of IHD in COPD patients only in elderly group. The vaccination rate of COPD patients with ischemic heart disease had higher vaccination rate. We found the influenza vaccination acceptance have increased significantly. However, this study could not explain whether the vaccine intervention helps to reduce ischemic heart disease without further research clinical trial investigations. Furthermore, if the severity and comorbidities of the disease further understood, it will provide information that is more comprehensive.

Acknowledgment

We would like to acknowledge this research which is sponsored in part by Department of Health, Executive Yuan, Taiwan under grants DOH101-TD -C-111-008.

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